[INCH-POUNDS]
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September 2, 1996
SUPERSEDING
MIL-P-83668A
April 8, 1988

COMMERCIAL ITEM DESCRIPTION

PLASTIC BOARD (FOR PACKAGING APPLICATIONS)

The General Services Administration has authorized the use of this commercial item description as a replacement for Type I (profile extrusion) and Type II (laminated corrugated) plastic board of MIL-P-83668 for all federal agencies.

1. SCOPE

This commercial item description covers plastic board primarily for use in the fabrication of shipping containers. Plastic Board may also be used in the fabrication of pads, sleeves, box liners, partitions, and cut shapes.

2. CLASSIFICATION

Plastic boards shall be of the following types:

Type I - Profile extrusion
Type II - Laminated corrugated

3. SALIENT CHARACTERISTICS

3.1 Materials. The plastic board shall be produced from a single material of uniform composition. The material shall be polyethylene, polypropylene, ethylene copolymer, or propylene copolymer. When the plastic board is available in either polyethylene, polypropylene or copolymers, the procuring activity

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AMSC N/A FSC 8135

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may specify the desired material. When the plastic board specified is intended for food applications, the material shall conform to 21 CFR Federal Food, Drug, and Cosmetic Act, Food Additive Amendment, Paragraph 121.2501 of Subpart F (see 4.2 and 7.2).

3.2 Color. Material shall be of the color specified.

3.3 Form. The finished plastic board shall be furnished in flat sheets. The sheets shall lay flat within 1/8 in. (3.2 mm) when tested in accordance with the following “Flatness Test”: Specimens 12 in. x 12 in. (304.8 mm x 304.8 mm) shall be cut from the middle of the sheets. Condition the specimens in accordance with 3.5. Place the specimens on a flat surface and measure the maximum vertical distance from the surface to the underside of the specimen.

3.4 Construction.

3.4.1 Type I. Type I plastic board shall be a profile extrusion with two facings separated by ribs.

3.4.2. Type II. Type II plastic board shall be in a laminated structure with two facings separated by a corrugating medium.

3.5 Test conditions for physical properties. Unless otherwise specified, specimens being tested for flatness (see 3.3), basis weight (see 3.6.1), thickness (see 3.6.2) and bending qualities (see 3.6.5) shall be conditioned for a minimum of 24 hours at 73.4°F ± 2°F (23°C ± 2°C) and 50 percent relative humidity (RH) ± 5 percent RH. Heat deflection (see 3.6.3)and Izod impact (see 3.6.4) test specimens shall be preconditioned for a minimum of 24 hours at 65°F to 85°F (18.3°C to 29.4°C).

3.6 Physical properties.

3.6.1 Basis weight. The basis weight shall not be less than the lower limit nor greater by more than 5 percent of the upper limit of the basis weight range specified (see 7.2) when tested in accordance with the following “Basis Weight Test”: With the exceptions specified in 3.5 and this paragraph, basis weight shall be determined in accordance with ASTM-D646. A minimum of five test units from the same plastic board sheet with minimum dimensions of 10 in. x 10 in. (254 mm x 254 mm) shall be used for a basis weight determination. The basis weight shall be determined for each of two sheets.

3.6.2 Thickness. The board thickness provided by the manufacturer shall be appropriate for the specified basis weight. No single thickness measurement shall vary by more than ± 10 percent from the manufacturer's designated thickness when tested in accordance with the following “Thickness Test”: With the exceptions specified in 3.5 and this paragraph, thickness shall be determined in accordance with TAPPI-T411. Five representative

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test specimens, with minimum dimensions of 3 in. x 5 in. (76.2 mm x 127 mm), shall be cut from each sheet of plastic board. The thickness of each test specimen shall be measured at five non-overlapping points. The thickness of five test specimens from each of two sheets shall be determined.

3.6.3 Deflection temperature at 66 psi (455.1 Kpa). The material from which the board is produced shall have a minimum deflection temperature of 150°F (65.6°C) at a fiber stress of 66 psi (455.1 KPa) when tested in accordance with the following "Deflection Temperature at 66 psi (455.1 Kpa) Test": Except as specified in 3.5 and this paragraph, the deflection temperature at a fiber stress of 66 psi (455.1 KPa) shall be determined in accordance with ASTM-D648. Test specimens shall be produced from the same material used in fulfilling the contract with molding conditions being those recommended by the manufacturer of the material. Test specimens shall be 1/4 in. x 1/2 in. x 5 in. (6.4 mm x 12.7 mm x 127 mm). Five determinations shall be conducted.

3.6.4 Notched Izod impact strength at 0°F (-17.8°C). The material from which the board is produced shall have a minimum notched Izod impact strength (i.e., energy per unit length of notch) of 0.7 ft-lb./in. of notch (0.10 Kg-m/cm of notch) at 0°F (-17.8°C) when tested in accordance with the following "Notched Izod Impact Strength at 0°F (-17.8°C) Test": With the exceptions specified in 3.5 and this paragraph, the impact resistance at 0°F (-17.8°C) shall be determined in accordance with Method D of ASTM-D256. Test specimens shall be produced from the same material used in fulfilling the contract with the molding conditions to be those recommended by the manufacturer of the material. Test specimens shall be 1/4 in. x 1/2 in. x 5 in. (6.4 mm x 12.7 mm x 127 mm). Three determinations shall be conducted.

3.6.5 Bending qualities. The plastic board shall show no visual surface breaks or cracks when tested in accordance with the following "Bending Qualities Test": Specimens, 12 in. x 12 in. (304.8 mm x 304.8 mm), shall be scored parallel and perpendicular to the direction of the ribs or corrugating medium. The test specimen shall be inspected for score line surface breaks or cracks. Cracks or breaks occurring at the intersection of score lines shall be disregarded. The specimen shall then be folded 180 degrees toward one facing and 180 degrees toward the other along the scoreline both parallel and perpendicular to the direction of the ribs or corrugating medium. The specimens shall be inspected for breaks or cracks after folding. The bending qualities of one test specimen from each of two sheets shall be determined.

3.6.6 Flute T-peel strength (Type II only). Type II plastic board, with basis weight of 140 pounds per 1000 square feet (63.5 Kg per 92.9 square meters) or less, shall have a minimum flute T-peel strength of 5 lb. (22.2 N). Plastic board, with basis weight greater than 140 pounds per 1000 square feet (63.5 Kg per 92.9 square meters), shall have a minimum flute T-peel strength

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of 35 lb. (155.7 N) Flute T-peel strength shall be determined in accordance with the following “Flute T-Peel Strength Test”:

3.6.6.1 Apparatus. A tension testing machine capable of applying a tensile load and having the following characteristics:

a. A loading range so that the maximum load on the specimen falls between 15 and 85 percent of the upper limit of the loading range.

b. A constant rate of movement between heads of 1 inch per minute (25.4 mm per minute).

c. The machine shall be equipped with suitable grips capable of clamping the specimens firmly without slippage during the test.

d. The machine shall be autographic with a chart that can be read in terms of inches of separation as one coordinate and applied load as the other coordinate.

e. The applied load as measured and recorded shall be accurate within ± 1 percent.

3.6.6.2 Test specimens. Five test specimens shall be cut from each of two plastic board sheets so that no portion of the test specimens is within 3 in. (76.2 mm) of any of the sheet edges. The test specimens shall be cut to the width necessary to include exactly 5 flute peaks on each facing. The five flute peaks on each facing shall form the only bonds of the flutes to the facings. The test specimens shall be 12 in. (304.8 mm) long. Using a sharp knife or razor blade, separate the facings at one end to a length of 3 in. (76.2 mm) by cutting through the center of the corrugations being careful not to damage the facings.

3.6.6.3 Test procedure. Clamp one of the facings, separated in the preparation of the test specimens, in the upper grip and the other facing in the lower grip of the tension testing machine aligning the specimen so that during the test, the load is applied along a line parallel to the specimen length. Apply the load at a constant rate of separation of grips on 1 inch per minute (25.4 mm per minute). During the test, make an autographic recording of load versus grip separation. Initially, the load will increase as tension builds in the corrugated medium, before the corrugations actually begin to tear. The actual start of tearing through the corrugations is reflected by a peak load. The load will decrease as the test continues due to the tear drifting towards one of the specimen facings. The recorded load will reflect a steady state when the corrugations begin peeling from the specimen facing. Measurement of peel strength shall be taken during the steady state peel and shall be continued for a minimum of 6 in. (152.4 mm) of separation (see FIGURE 1), or until one of the facings ruptures. The recorded load, during steady state, or when one of the facings ruptures

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shall not be less than the value specified in 3.6.6. The flute T-peel strength of five test specimens from each of two sheets shall be determined.

3.7 Dimensions. Sheet size, length and width, shall be as specified by the procuring activity (see 7.2).

3.7.1 Dimensional tolerances. The dimensional tolerances shall be minus 0, +1/2 in. (-0, +12.7 mm) for both length and width of the specified sheet size.

3.8 Workmanship. The plastic board shall contain no foreign matter except dust or dirt that adheres due to the tendency of plastics to accumulate electric charges. There shall be no scuffs, blisters, tears, holes, cuts, marks, or splices. Trimmed edges must be perpendicular and free from burrs. Dirt or dust particles, which adhere to the plastic board because of the tendency of plastics to accumulate electric charges, shall be disregarded in the examination for cleanliness.

4. REGULATORY REQUIREMENTS

4.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 When the plastic board specified is intended for food applications, the material shall conform to 21 CFR Federal Food, Drug, and Cosmetic Act, Food Additive Amendment, Paragraph 121.2501 of Subpart F (see 7.2).

(Applications for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

5. QUALITY ASSURANCE PROVISIONS

5.1. Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market Acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

5.2.1 The company, and its immediate predecessors, producing this material shall have been producing uninterrupted production of a product meeting the requirements of this CID for at least three years and has not been barred from selling to the

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government in the last five years, or until founded, whichever is less.

5.2.2 Distributors purchasing products from a manufacturer shall guarantee that the manufacturer meets the requirement stated above. Only the product from one manufacturer shall be used to fulfill a single purchase order.

6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Intended use. Plastic board is primarily intended for use in the fabrication of reusable boxes. In addition, plastic board may be used for those packaging applications requiring a container material whose properties are unaffected by exposure to water. Plastic board may also be used in the fabrication of pads, sleeves, box liners, partitions, and cut shapes. Plastic board, like most other plastic materials, exhibits the tendency to accumulate and store large quantities of electric charges. It is therefore not recommended for packaging flammable and explosive materials or materials subject to damage from electrostatic charges unless the package or pack has been designed to prevent such hazards.

7.2 Ordering data. Purchasers should include the following information in procurement documents:

a. Title, number, and date of this specification.

b. Quantity (number of sheets).

c. Type (see 2.1 and 3.4).

d. Specify the raw material (polyethylene, polypropylene, or copolymer) if a specific material is required (see 3.1).

e. Specify if food grade board is required (see 3.1).

f. Specify color (see 3.2)

g. Basis weight (see 3.6.1). It may be desirable to specify a basis weight range such as 160 to 170 pounds per 1000 square feet (72.6 to 77.1 Kg per 92.9 square meters).

h. Length and width of sheet (see 3.7).

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7.3 Keywords.

Plastic corrugated, Box liners, Packaging, Pads, Partitions, Plastic Panel, Box Panel, Plastic board, Plastic Corrugated Board, Sleeves

7.4 Referenced Documents.

7.4.1 Government documents. Copies of specifications, standards, handbooks, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

Title 21 Code of Federal Regulations Federal Food, Drug, and Cosmetic Act, Food Additive Amendment

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis from the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuance thereof.)

7.4.2 Non-Government documents.

7.4.2.1 ASTM standards. ASTM standards are available from the American Society for Test and Materials, 100 Barr Harbor Drive, West Conshohocken PA 19428-2959.

ASTM-D256	Standard Test Methods for Determining the Pendulum Impact Resisteance of Notched Specimens of Plastics (DoD Adopted)
ASTM-D646	Standard Test Method for Grammage of Paper and Paperboard
ASTM-D648	Standard Test Method for Deflection Temperature of Plastics Under Flexural Load (DoD Adopted)

7.4.2.2 TAPPI standards. Applications for copies of TAPPI standards by serial designation and title should be made to Technical Association of the Pulp and Paper Industry, PO Box 105113, Atlanta GA 30348.

TAPPI-T411	Thickness (caliper) of Paper, Paperboard, and Combined Board

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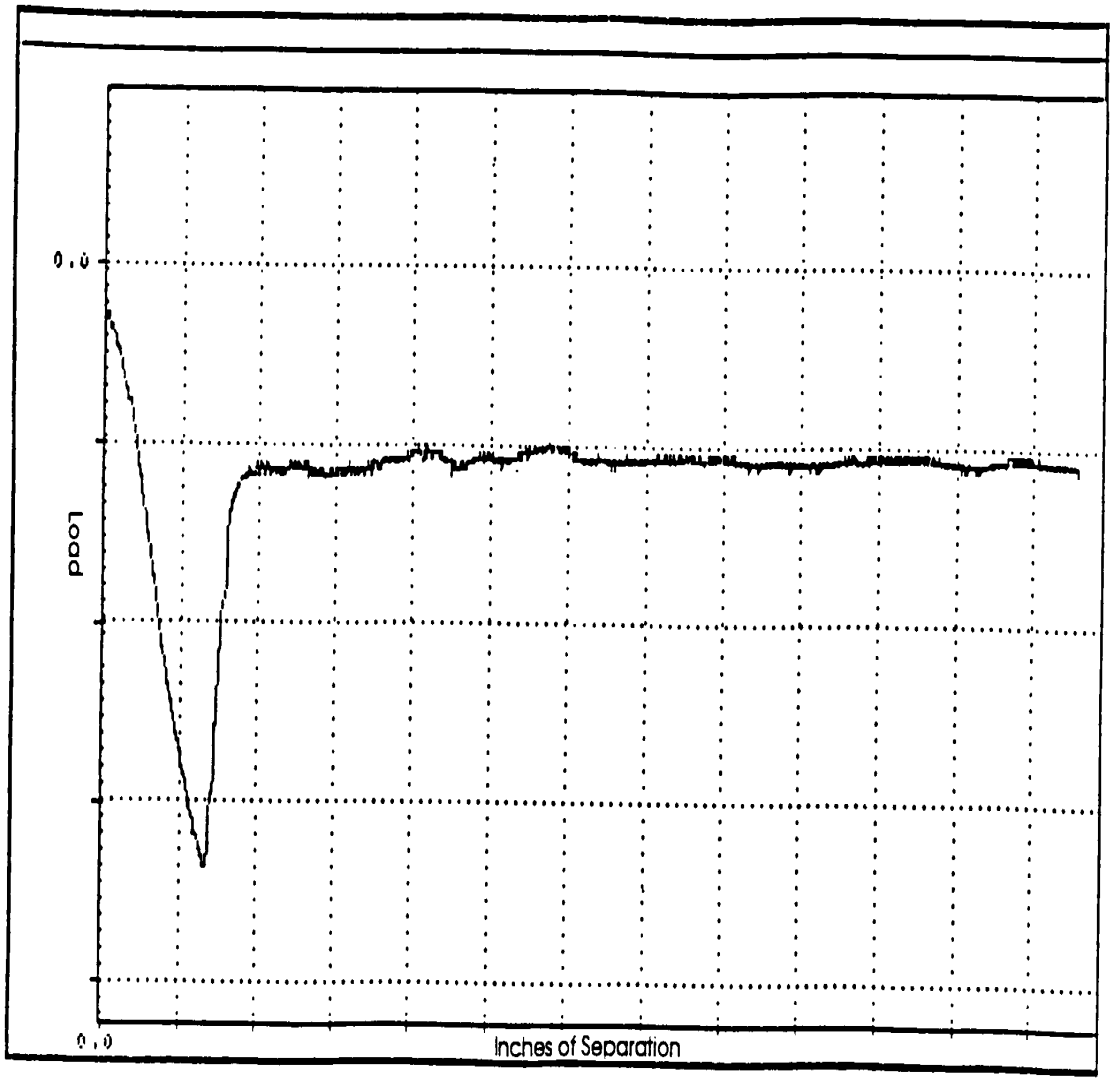

FIGURE 1. Flute T-Peel Strength Test - Load Versus Inches of Separation.

Custodian:

Army - SM
GSA - 2FYEE
Navy - SA

Preparing activity:

Air Force - 69

(Project 8135-0700)

Review activities:

Air Force - 11, 70, 71, 80, 84, 99

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